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Whole-genome sequencing and genomic characterization of a novel multi-drug resistant *esxA*-positive *Staphylococcus haemolyticus* DUEML1 (ST-184) isolated from a respiratory infection case: insights from panresistome analysis

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Abstract

Background *Staphylococcus haemolyticus* is a coagulase-negative staphylococcal species and an opportunistic pathogen associated with hospital-acquired infections. The aim of this study was to use whole-genome sequencing (WGS) to characterize a novel multidrug-resistant (MDR) *S. haemolyticus* strain, DUEML1 (ST-184), isolated from a respiratory infection case in Bangladesh, and to place its resistome and virulence features in the context of global *S. haemolyticus* isolates.

Methods The isolate was obtained in pure culture from the tracheal aspirate of a 51-year-old male patient with respiratory infection, suggesting it was the primary causative agent. WGS was the primary method to analyze the genome of the isolated strain and subsequent in-silico analyses were performed to identify antimicrobial resistance genes, virulence factors, plasmid-associated genes, mobile genetic elements (MGEs), and prophages. Comparative pan-resistome analysis was conducted using 694 publicly available *S. haemolyticus* genomes retrieved from NCBI.

Results The isolate exhibited in vitro resistance to levofloxacin, ciprofloxacin, tetracycline, doxycycline, gentamicin, and trimethoprim-sulfamethoxazole, as determined by the disc diffusion test, and demonstrated the capacity for biofilm formation. Several antimicrobial resistance genes (ARGs) such as *fosBx1*, *mgrA*, *norC*, *sdrM*, *sepA* and two virulence genes, including *esxA* and *cap8G* were detected. To our knowledge, this is the first report of an *esxA*-positive *S. haemolyticus* isolate, recovered from a respiratory infection case in Dhaka, Bangladesh. Two plasmid-associated genes *repUS23* and *repUS46* were detected. Further analyses predicted 63 horizontal gene transfer (HGT) events and identified 147 MGEs, including integration/excision elements, recombination and repair-associated genes, and prophage-associated regions. With a new variant of *arcC* allele (*arcC-38*), the isolate was assigned to a novel ST-184.

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The PathogenFinder predicted a 93% probability that ST-184 is a human pathogen. A comparative analysis of 694 genome sequences identified a wide variety of ARGs, virulence factors, and plasmids in *S. haemolyticus* isolates from 35 different countries.

Conclusion This study provides the first genomic characterization of a novel *S. haemolyticus* ST-184 isolate from Bangladesh, highlighting its multidrug-resistant nature and virulence potential. A limitation of this work is the lack of clinical treatment outcome data. Future research should include large-scale genomic surveillance to strengthen our understanding of the genomic architecture of *S. haemolyticus*.

Keywords *Staphylococcus haemolyticus* ST-184, Multi-drug resistance, Respiratory tract infection, Biofilm, Plasmid-associated genes, Virulence factors, Resistance genes

Background

Staphylococcus haemolyticus is a coagulase-negative *staphylococcus* (CoNS) species increasingly recognized as an opportunistic pathogen in nosocomial infections [1]. Commonly residing as a commensal on human skin and mucous membranes, *S. haemolyticus* has been implicated in bloodstream infections, endocarditis, respiratory tract infections, urinary tract infections, and infections associated with indwelling medical devices, particularly among immunocompromised patients [1, 2]. The ability of this species to develop resistance to multiple antibiotics, form biofilms, and acquire mobile genetic elements contributes to its persistence in the clinical environment [3].

The resistance profile of *S. haemolyticus* is alarming: over 80% of clinical strains exhibit methicillin resistance, with increasing tolerance to glycopeptides, fluoroquinolones, and aminoglycosides [4, 5]. Similarly, the genomic landscape of *S. haemolyticus* is marked by a high prevalence of resistance determinants. Efflux pumps (e.g., *qacG*, *norC*, *sdrM*) and modifying enzymes (e.g., *aac(6')-aph(2")*, *fosBx1*) confer resistance to antiseptics, fluoroquinolones, and aminoglycosides [5, 6]. Recently, in Bangladesh, antimicrobial resistance (AMR) has emerged as a critical public health issue, particularly in the hospitals where the reports suggest that multidrug-resistant (MDR) strains of *S. haemolyticus* are increasingly being isolated from patients with nosocomial infections, posing a huge challenge to the therapeutic methods [7]. A study in Dhaka indicated that *S. haemolyticus* was among the most frequently isolated CoNS species from clinical samples, with a high prevalence of resistance to fluoroquinolones, aminoglycosides, and tetracyclines [8]. These findings align with global trends, where MDR *S. haemolyticus* isolates have been reported in hospital environments in Europe, the Americas, and Asia, emphasizing the necessity of genomic surveillance to track the evolution and dissemination of resistance determinants [9]. Additionally, these results highlight the urgent need to explore novel antimicrobial agents and alternative therapeutic strategies to combat multidrug-resistant strains [10].

One of the prominent attributes of *S. haemolyticus* is the capacity to form biofilm, which provides a protective barrier that enhances bacterial survival in hostile environments, including exposure to antibiotics and the ability to form biofilms on medical devices such as catheters and prosthetic implants significantly contributes to chronic infections and therapeutic failures [11]. Several genes play crucial roles in biofilm formation, with the *ica* operon being one of the most well-characterized determinants [11]. Critically, biofilms not only shield bacteria from antibiotics but also serve as hotspots for the exchange of AMR and virulence genes through horizontal gene transfer (HGT) mechanisms such as transformation, transduction, and conjugation [12, 13]. In *S. haemolyticus*, the exchange of AMR and virulence genes occurs primarily through HGT, facilitated by mobile genetic elements (MGEs) including plasmids, transposons, and prophages [14]. Importantly, horizontal gene transfer is not restricted within species but can also occur across *staphylococcal* species. For example, mupirocin resistance plasmids have been shown to transfer from *S. haemolyticus* to *S. aureus*, demonstrating the capacity for interspecies exchange of clinically relevant genes [15]. Virulence factors such as *esxA*, which are well-characterized in *S. aureus* but have not previously been reported in *S. haemolyticus*, may therefore be acquired through such mechanisms. In *S. aureus*, EsxA is secreted by the ESAT-6-like secretion system (ESX-1), where it contributes to immune evasion and abscess formation [16]. EsxA forms a complex with EsxB, promoting the release of intracellular *S. aureus* from host epithelial cells and facilitating bacterial dissemination during infection. The transfer of such virulence genes to *S. haemolyticus* could pose a significant threat by enhancing its pathogenic potential and clinical relevance.

The identification of specific sequence types (STs) within *S. haemolyticus* populations provides valuable insights into the epidemiology and pathogenic potential of this opportunistic pathogen. For instance, a 2022 study in Taiwan identified sequence type 42 (ST42) as a dominant clone linked to elevated disease severity in burn patients, underscoring the clonal expansion of high-risk lineages [17]. Similarly, a study in India revealed 24 novel sequence types among *S. haemolyticus* isolates from various infections, highlighting the genetic diversity and adaptive potential of this species [18]. The ST can be determined by using whole-genome sequencing (WGS). The WGS is a powerful method for determining the complete or nearly complete DNA sequence of an organism's genome in a single process. This approach provides a comprehensive understanding of an organism's genetic makeup [19]. Recent advancements in bacterial WGS have solidified its status as a widely adopted technique across various fields, enabling detailed analyses of bacterial pathogens and offering crucial insights into antibiotic resistance, molecular epidemiology, and virulence, which not only enhance our understanding of its biology but also inform potential strategies to counteract resistance to existing treatments [20]. In the face of rising concerns over MDR, extensively drug-resistant (XDR), and pan-drug-resistant (PDR) bacteria, WGS and metagenomic sequencing have become indispensable tools in infectious disease research. This wealth of genomic data is essential for developing effective strategies to mitigate the serious threats posed by these highly resistant pathogens [21, 22]. In previous reports, WGS has uncovered extensive genetic diversity in *S. haemolyticus*, driven by recombination, phage integration, and MGEs [2, 17]. However, more region-specific studies are needed to complement global insights and provide a clearer understanding.

In Bangladesh, although *S. haemolyticus* has frequently been associated with various types of infections there is limited genomic information available about this pathogen in the region [8]. The lack of genomic information suggests there might be novel sequence types yet to be reported. The absence or scarcity of genomic information in a particular region halts genomic surveillance efforts. Hence, there is a need for comprehensive genomic studies to identify and characterize emerging strains, track their evolutionary dynamics, and assess their potential public health impact. In this study, we have reported a novel MDR *S. haemolyticus* DUEML1 (ST-184) from Bangladesh. Here, along with analyzing the antibiotic susceptibility pattern and biofilm-forming capacity, we have conducted whole-genome sequencing to investigate the isolate's virulence genes, antimicrobial resistance genes, mobile genetic elements, biosynthetic gene clusters, and tried to explore their potential associations and roles in pathogenesis. Besides, we have carried out panresistome analysis and comparative analysis of virulence and plasmid-associated genes to capture the overall trend. These findings highlight the need for genomic surveillance to inform infection control policies and antimicrobial therapies in resource-limited settings.

Methodology

The overall workflow of the study is illustrated in Fig. 1.

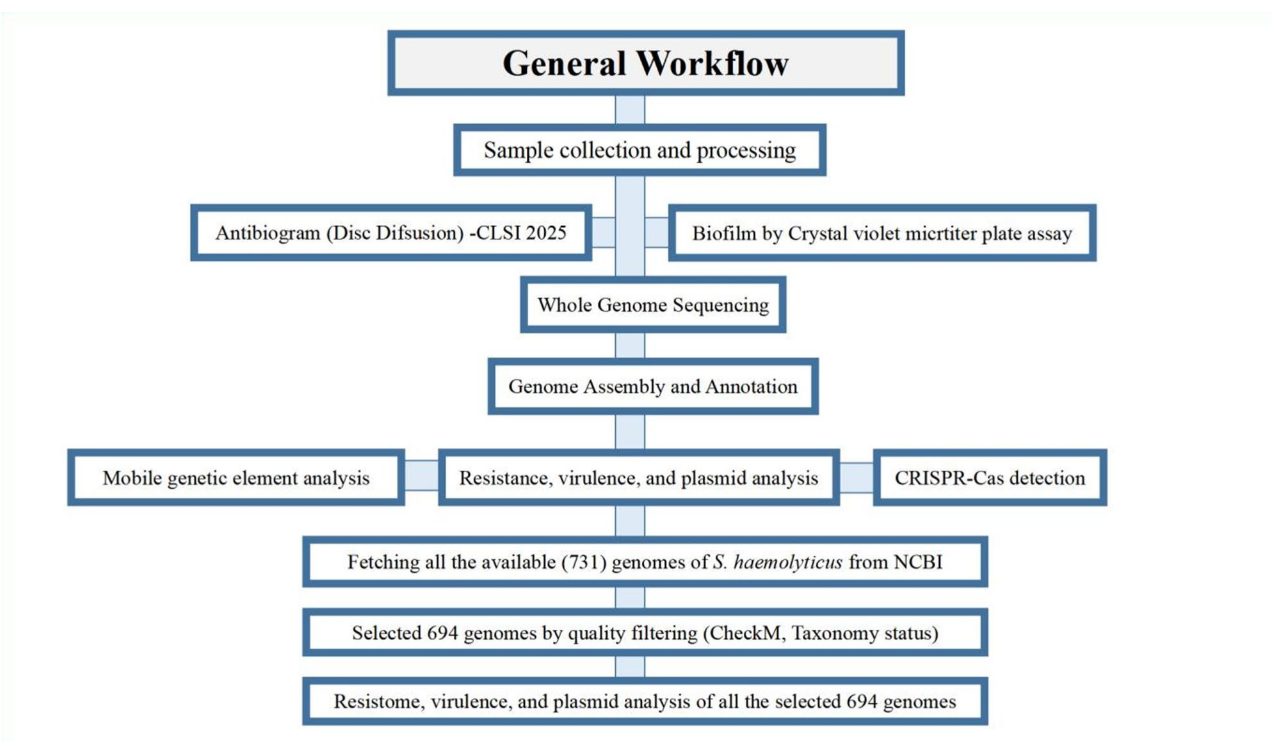

Fig. 1 Workflow of the study methodology

Sample collection, processing and antibiotic susceptibility testing

The sample was collected from the tracheal aspirate of a 51-year-old male patient with a respiratory infection at a hospital in Dhaka. Medical professionals collected the sample using a sterile suction catheter under aseptic conditions. It was immediately transported to the Environmental Microbiology Laboratory (LAB 206), Department of Microbiology, University of Dhaka, in a cooler with ice packs (maintained below 4 °C) and processed within a few hours. Initially, the tracheal aspirate was inoculated into tryptic soy broth (TSB) supplemented with 6.5% NaCl and incubated for 18–24 h at 37 °C to enrich for *staphylococci*. Following enrichment, cultures were streaked onto Mannitol Salt Agar (MSA) plates to selectively isolate members of the *Staphylococcaceae* family. The predominant colonies screened out from MSA plates were further cultured for isolating pure colonies. Several biochemical tests were carried out along with observing cultural and morphological characteristics for the presumptive identification of the isolate. Antibiotic susceptibility testing (AST) was carried out on the isolates by using Mueller-Hinton agar by the Kirby-Bauer disk diffusion method [23]. The results were categorized as sensitive, intermediate, or resistant according to the Clinical and Laboratory Standards Institute's (CLSI) M100-Ed35:2025 guidelines. The following antibiotic discs were used in the assay: Levofloxacin, Ciprofloxacin, Trimethoprim-sulfamethoxazole, Tetracycline, Doxycycline, Chloramphenicol, Linezolid, Azithromycin, Gentamicin, and Cefoxitin. The isolate was considered MDR if it showed resistance to at least one antibiotic in three or more different antimicrobial categories [24, 25]. The biofilm formation ability was determined by microtiter dish biofilm formation assay using 96-well polystyrene plates [26].

Whole genome sequencing, assembly and identification

For whole genome sequencing, genomic DNA was extracted using the QIAamp DNA Mini Kit (QIAGEN, Hilden, Germany) according to the manufacturer's protocol. The DNA concentration was measured using the Colibri Microvolume Spectrometer (Titertek-Berthold, Berthold Detection Systems GmbH, Bleichstrasse, Pforzheim, Germany), while DNA integrity was verified by electrophoresis on 0.8% (w/v) agarose gels. At the icddr, b Genomics Centre (iGC), the whole genome sequencing of the target isolate was carried out using the Illumina NextSeq sequencing technology. The quality of the raw FASTQ files were evaluated using FastQC v0.12 [27]. To remove low-quality bases and adapter sequences, reads were processed using Trimmomatic v0.39 [28]. Reads with an average quality score below 30 and a length shorter than 50 base pairs were discarded. After trimming, the good-quality reads were subjected to de novo assembly using SPAdes v4.0.0 (Species Prediction and Diversity Estimation) [29]. The assembly quality was evaluated using CheckM [30]. The whole genome sequence-based identification of the isolate was carried out through KmerFinder tool of Center for Genomic Epidemiology [31]. ANI (Average Nucleotide Identity) based identification was performed using fastANI [32], with the *S. haemolyticus* ATCC 29,970 strain serving as the reference for ANI-based identification. The sequence type (ST) of the isolate was identified using a multilocus sequence typing (MLST) scheme based on seven housekeeping genes (*arcC*, *riboseABC*, *cfxE*, *hemH*, *leuB SH1431*, and *SH1200*) [33]. The allelic profile of the isolate was compared to the profiles in the *S. haemolyticus* PubMLST database [34].

Genome annotation, genome mapping, and biosynthetic gene cluster analysis

The assembled sequence underwent annotation through Prokka, Bakta, and Prokaryotic genome annotation pipeline (PGAP) of NCBI [35–37]. The circular genome map was constructed using Proksee server [38]. The biosynthetic gene clusters were determined through the antiSMASH7.0 tool [39].

Investigation of antimicrobial resistance genes, virulence factors, mobile genetic elements and pathogenicity

Resistance genes were screened using Abricate using ResFinder 3.1, CARD (Comprehensive Antibiotic Resistance Database), and NCBI [40–42]. Similarly, plasmids and virulence genes were screened using plasmidFinder and VFDB, respectively [43, 44]. The pathogenicity of the bacterial isolate towards human hosts was predicted using the PathogenFinder tool [45].

Investigation of mobile genetic elements, prophage, insertion sequences (IS), antiviral defense systems

Mobile genetic elements were explored through mobileOG-db65 [46]. Besides, prophage investigation in the target bacterial genome was carried out through the Phigaro tool [47]. Horizontally acquired genomic regions were identified using AlienHunter [48]. The ISfinder tool was used to identify the insertion sequences (IS) [49]. The significance threshold for the analysis was set at an E-value < 1e-10. The CRISPR arrays and their associated Cas proteins were detected using CRISPR/Cas Finder [50].

Comparative analysis of resistance, plasmids and virulence factors

All available genomes of *S. haemolyticus* were retrieved from the National Center for Biotechnology Information (NCBI) genome database. The genomes were subsequently filtered based on assembly quality (checkM completeness > 95%) and taxonomy status (Ok). The metadata for selected genomes was retrieved using the Entrez Programming Utilities (E-utilities) provided by the NCBI [51]. The sequences were then screened for resistance genes, virulence factors, and plasmids as described previously.

Results and discussion

Multidrug resistance and biofilm forming ability

The target isolate showed complete resistance to Levofloxacin, Ciprofloxacin, Tetracycline, Doxycycline, Gentamicin, and Trimethoprim-sulfamethoxazole, which belongs to four different categories of antibiotics. The isolate was sensitive to the rest of the antibiotics tested. Based on this resistance profile, the isolate qualifies as MDR, as it demonstrated non-susceptibility to at least one agent in three or more antimicrobial categories. Based on sensitivity to cefoxitin, the isolate can be referred to as methicillin sensitive *S. haemolyticus* (MSSH). While being MSSH, the isolate still poses risk to immunocompromised individuals with its MDR profile. Since it can serve as a source of drug resistance, we evaluated biofilm-forming ability of the isolate based on microtiter biofilm formation assay. Biofilm formation is one of the acknowledged strategies that helps bacteria to endure harsh conditions by developing a protective matrix [52]. In the assay, the isolate showed biofilm-forming capacity and was categorized as a weak biofilm former with a mean Optical Density (OD) of 0.183 and a mean OD/OD_{cut} ratio of 1.45. The ability of the isolate to form biofilms, even as a weak biofilm former, carries significant clinical implications, especially considering that the isolate was recovered from a tracheal aspirate sample, where biofilm formation could contribute to persistent infections and antibiotic tolerance. Inoculation studies in mice showed that *S. haemolyticus* biofilms cause interstitial pneumonia and trigger inflammatory responses (e.g., TNF-α, MIP-2) [53]. Even low bacterial concentrations induce lung injury, with biofilm formation enabling survival despite immune clearance efforts [54]. With such a multidrug resistance profile and biofilm-forming ability, novel therapeutic strategies, such as the use of biofilm-disrupting agents, phage therapy, and novel antimicrobial agents should be explored to complement traditional treatments.

Whole genome sequence analysis, annotation and identification

Genomic reports of *S. haemolyticus* from Bangladesh are scarce, with only a few reports [7]. A recent study reported the draft genome sequences of two multi-drug-resistant *S. haemolyticus* strains, SAC2 and SAC7, isolated from clinical samples in Dhaka [7]. Genomic analyses revealed the presence of resistance genes, virulence factors, and toxin-antitoxin systems, providing insights into the genomic content and epidemiology of *S. haemolyticus* in Bangladesh. Another study sequenced the genome of *S. haemolyticus* strain 173 N-3, isolated from a nasal swab of a healthy individual [55]. The genomic analysis identified antibiotic resistance genes, including *msrA* and *mphC*, associated with macrolide resistance, highlighting the potential for resistance gene acquisition even in asymptomatic carriers. These studies represent only a few of the available reports on *S. haemolyticus* from Bangladesh.

The lack of well-characterized whole genomes from Bangladesh, combined with the multidrug-resistance profile of this isolate, prompted us to sequence its genome. The whole genome sequence-based identification using KmerFinder further confirmed the isolate as *S. haemolyticus.* Moreover, the ANI percentage with another *S. haemolyticus* ATCC 29,970 was 97.52%, which also validates the identification of the isolate as *S. haemolyticus*. Afterward, the isolate was designated as "*Staphylococcus haemolyticus* DUEML1". The sequence typing result revealed the isolate to be of a new sequence type with similarity to ST −7, ST-89, and ST-149. A novel variant of the housekeeping gene *arcC* was identified and submitted to the PubMLST database. With the new *arcC* allele (*arcC-38*), the isolate was assigned to a new sequence type, ST-184. The identification of a novel sequence type suggests possible genomic diversification within *S. haemolyticus*, which may contribute to its adaptation and persistence in clinical settings. This report can also serve as a reference for ST-184 in future studies. After sequence type identification, the assembled whole genome sequence was deposited in GenBank of NCBI under the accession number JBLFEB000000000. The analysis of the genome indicated good quality of the genome considering the completeness (99.25%) and low contamination (0.08%). The assembled genome of *S. haemolyticus* consisted of 13 contigs, with a total length of 2,479,125 base pairs and an average GC content of 33%. The genome size and GC content of the sequenced isolates aligned well with the established genomes of *S. haemolyticus*. Following the de novo assembly and annotation process, the CDS ratio of 0.96 indicates a high-quality and successful assembly. This conclusion is supported by the fact that the CDS ratio aligns with the standard range of 0.8 to 1.2. The genome mapping and annotation revealed significant features of the genome which are listed in (Table 1; Fig. 2).

Investigation of antimicrobial resistance genes, virulence factors, mobile genetic elements and pathogenicity

The annotation revealed a diverse array of genes, necessitating further analysis to elucidate their roles in the

Table 1 General features of the assembled whole genome

Features (Annotation)	**PGAP**	**Prokka**	**Bakta**
Genes	2472		
CDS	2402	2454	2391
Protein Coding	2376		
tRNAs	59	59	59
rRNAs	4	4	4
ncRNAs	4		38
Features (Assembly)			
Genome Size	2,479,125 bp	Completeness (CheckM)	99.25%
Contigs	13	Contamination (CheckM)	0.08%
GC Content	33%	CDS ratio	0.96
Contig N50	424.9 kb	Genome coverage	149.4x
Contig L50	3		

Table 2 List of resistance and virulence genes in the genome of *S. haemolyticus* DUEML1 (ST-184)

Feature	**Gene**	**Drug Class**	**Resistance Mechanism**
Antimicrobial Resistance Genes	*FosBx1*	phosphonic acids (fosfomycin)	Antibiotic inactivation
	norC	fluoroquinolone antibiotics (e.g., ciprofloxacin, levofloxacin, norfloxacin)	Antibiotic efflux
	sdrM	fluoroquinolones (ciprofloxacin, norfloxacin), macrolides (erythromycin), tetracyclines, and aminoglycosides	Antibiotic efflux
	sepA	Disinfectants and antiseptics	Antimicrobial efflux
	mgrA	β-lactams (penicillins, cephalosporins), fluoroquinolones, and aminoglycosides	Regulate expression of multiple resistance genes
Virulence genes	Gene	Product	Function
	esxA	Type VII secretion system protein secreted protein EsxA	Host cell membrane disruption and cytolysis
	cap8G	Capsular polysaccharide synthesis enzyme Cap8g	Protect the bacteria from host immune defenses

pathogenesis and resistance of the newly identified *S. haemolyticus* ST-184. Here, we have characterized the novel ST-184 by examining its resistance determinants, virulence factors, mobile genetic elements, pathogenicity-associated genes, and prophages to gain a comprehensive understanding of its genomic architecture. The antimicrobial resistance genes investigation revealed that *S. haemolyticus* contained five genes, including *fosBx1*, *mgrA*, *norC*, *sepA*, and *sdrM* responsible for conferring resistance to many antimicrobials (Table 2; Fig. 3c). The *FosBx1* gene, which encodes a fosfomycin-inactivating enzyme, confers resistance to fosfomycin by modifying the drug, rendering it ineffective [56]. The efflux pump genes *norC*, *sepA*, and *sdrM* contribute to multidrug resistance by actively extruding antimicrobials from the bacterial cell [57–59]. *NorC*, a member of the Major Facilitator Superfamily (MFS), has been associated with resistance to fluoroquinolones, while *sdrM*,

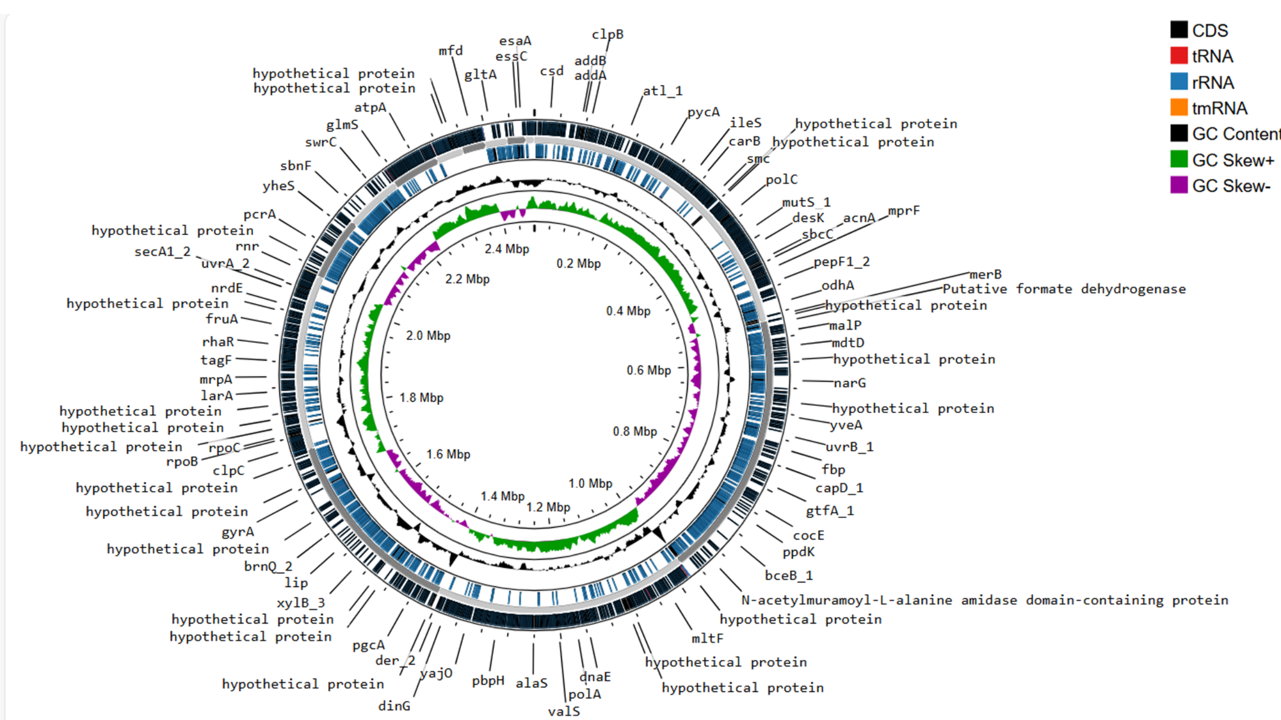

Fig. 2 Whole genome mapping of *S. haemolyticus* DUEML1

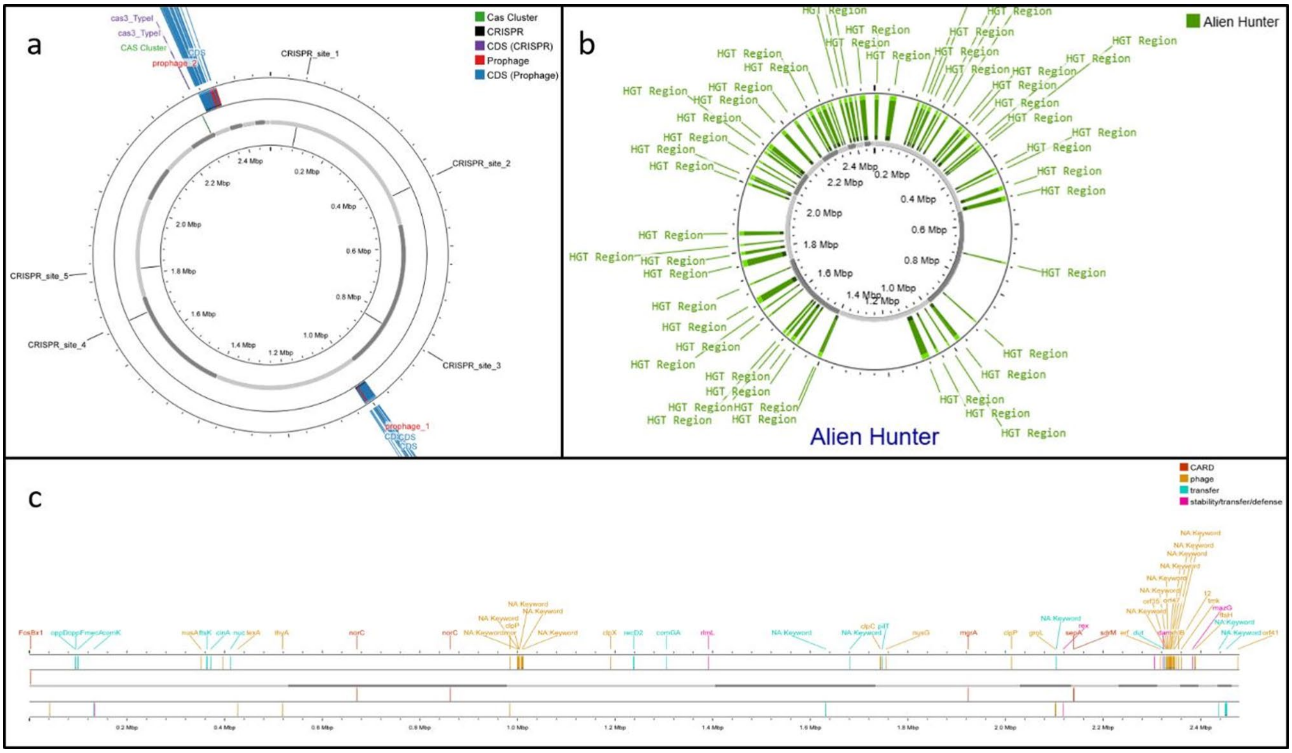

Fig. 3 Distribution of CRISPR-Cas, prophage regions, horizontal gene transfer events, antimicrobial resistance genes, mobile genetic elements associated genes in the genome of *S. haemolyticus* DUEML1. **a** Mapping of CRISPR-Cas and prophage regions. **b** Mapping of horizontal gene transfer events. **c** Mapping of antimicrobial resistance genes and mobile genetic elements

an ATP-binding cassette (ABC) transporter, is known to mediate resistance to a wide range of antimicrobial agents, including macrolides and aminoglycosides [57, 58, 60]. The *sepA* is part of the small multidrug resistance (SMR) family of efflux pumps [58]. It actively exports disinfectants and antiseptics, reducing their intracellular concentrations [61]. Additionally, the global regulator *mgrA* was detected, a transcriptional regulator known to control multiple resistance mechanisms, including the expression of efflux pumps and biofilm formation [61, 62]. MgrA plays a crucial role in reducing susceptibility to β-lactams, vancomycin, and other antimicrobial agents by altering bacterial membrane permeability and increasing resistance to host immune responses [61, 62].

When we analyzed the virulence profile by comparing the genome with the VFDB database, we found two virulence genes, namely *esxA* and *cap8G*, associated with the bacterium (Table 2). To the best of our knowledge, no published reports have described the presence of the *esxA* gene in the *S. haemolyticus* genome. The presence of *esxA* in the novel *S. haemolyticus* ST-184 suggests a potential acquisition through horizontal gene transfer, possibly mediated by mobile genetic elements such as plasmids or bacteriophages. The *esxA* gene in *S. aureus* encodes a secreted protein involved in immune evasion and cellular damage [17]. It is part of the type VII secretion system, contributing to the bacterium's ability to survive and persist within host tissues by disrupting immune responses [17]. In *S. aureus*, *esxA* and *esxB* typically function together as part of the type VII secretion system [17, 63]. However, in DUEML1, only *esxA* was detected, while *esxB* was absent. The presence of *esxA* without *esxB* may indicate a partial or divergent type VII secretion system in *S. haemolyticus*, the functional implications of which require further investigation. While the *esxA* allows the bacterium to disrupt the immune system, the capsule produced by *cap8G* prevents phagocytosis by immune cells, aiding the bacteria in establishing persistent infections in the respiratory tract of the immunocompromised individuals [64, 65]. The presence of these virulence determinants suggests that ST-184 may have a heightened pathogenic potential. PathogenFinderanalysis further supports this, predicting a 93% likelihood that it is a human pathogen, with 66 matches to known pathogenic families and no associations with non-pathogenic ones.

The pathogenicity, virulence factors, antimicrobial resistance genes in ST-184 may have been acquired or may become a source of reservoir through the dissemination of plasmids and other mobile genetic elements (MGEs). The analysis of plasmid-associated genes using PlasmidFinder identified two replication-associated genes within plasmid SAP099B: *repUS23*, encoding the replication initiator protein A (RepA), and *repUS46*, encoding a plasmid replication protein. Both of these plasmid-associated genes have previously been reported in *S. aureus* [66, 67]. The *repA* (*repUS23*) gene of SAP099B belong to the replicase type *rep_N* and the *repUS46* belong to the replicase type *rep_2* [66]. The RepA_N and Rep_2 both families consist of replication initiator proteins that recognize and bind specific sequences at the plasmid's origin to facilitate DNA replication initiation [68].

The MGEs were identified by analyzing with mobileOG-db, where we identified 147 MGEs in the genome (Fig. 3a). These included 10 integration/excision elements, 76 related to replication, recombination, and repair, 38 phage-associated elements, 7 involved in stability, transfer, and defence, and 16 associated with genetic transfer. Among Phage-associated elements, genes related to phage-mediated lysis, such as *nfuA*, were identified, suggesting a mechanism for phage-driven gene transfer [69]. These prophage-associated regions may facilitate gene mobilization, contributing to genetic diversity and the acquisition of new traits [70]. Replication-associated genes, including *addA* and *addB*, were identified within regions predicted to be involved in DNA recombination and repair [71, 72]. These genes play a crucial role in facilitating genetic exchange by enabling homologous recombination, which may contribute to genome rearrangements and adaptation to environmental stressors [72, 73]. Their presence indicates active recombination hotspots within the *S. haemolyticus* genome. Additionally, genes linked to competence-based transfer mechanisms, such as *oppD*, were detected. These genes are involved in DNA uptake and processing,

Table 3 List of insertion sequences detected in the genome of *S. haemolyticus* DUEML1 (ST-184)

Sequences producing significant alignments	IS Family	Group	Origin	Score (bits)	E. value
ISSha1	ISL3		*Staphylococcus haemolyticus*	464	8e-128
ISSep3	IS200/IS605	IS200	*Staphylococcus epidermidis*	325	4e-86
IS1469	IS200/IS605	IS200	*Clostridium perfringens*	95.6	8e-17
ISBsph2	IS200/IS605	IS200	*Bacillus sphaericus*	87.7	2e-14
ISBsph1	IS200/IS605	IS200	*Bacillus sphaericus*	87.7	2e-14
IS657	IS200/IS605	IS200	*Bacillus sphaericus*	81.8	1e-12
MICBce6	IS4	IS231	*Bacillus cereus*	73.8	3e-10

a key aspect of natural transformation [74]. The presence of these genes suggests that *S. haemolyticus* ST-184 may possess the ability to acquire extracellular DNA, further enhancing its adaptability. The *S. haemolyticus* biofilms primarily rely on proteins and extracellular DNA (eDNA) rather than polysaccharide intercellular adhesin (PIA). The presence of the *oppD* gene may contribute to the biofilm-forming ability of the ST-184 strain. Additionally, the Alien Hunter analysis predicted 63 genes as potential horizontal gene transfer (HGT) events, indicating the presence of mobile genetic elements that may have been acquired from other organisms (Fig. 3b). This suggests possible adaptation through gene exchange, which could contribute to functions like antibiotic resistance or virulence.

Investigation of prophage, insertion sequences, and and antiviral defense mechanism

To further investigate the phage elements in the genome, investigation of the prophage regions was investigated by PHIGARO (Prophage Identification and Genomic Analysis for Organisms). The analysis identified two distinct prophage regions. Prophage Region 1 spans 28,959 bp and contains 24 genes associated with the Siphoviridae family, a group of temperate bacteriophages characterized by their long, non-contractile tails (Fig. 3a). Prophage Region 2 is 40,985 bp in length and harbors 62 prophage-related genes, also belonging to the Siphoviridae family (Fig. 3a). Further analysis using ISfinder revealed the presence of multiple insertion sequence (IS) elements, demonstrating a diverse array of mobile genetic elements within the sequence (Table 3). Significant alignments (E-value < 1e-10) were observed for IS elements belonging to three families, including ISL3 (ISSha1), IS200/IS605 (ISSep3, IS1469, ISBsph2, ISBsph1, IS657), and IS4 (MICBce6). The insertion sequences identified in this study originate from a wide range of bacterial species, including *S. haemolyticus*, *S. epidermidis*, *Clostridium perfringens*, *Bacillus halodurans*, *Bacillus cereus*, and *Bacillus sphaericus* (Table 3). These IS elements, originating from diverse sources, likely play a significant role in genome plasticity and evolution. Additionally, the CRISPR/Cas Finder analysis using CRISPRCasFinder identified five CRISPR arrays and two Cas-associated elements (both Cas3) in the genome (Fig. 3a). The identification of a Cas3 element, characteristic of a Type I CRISPR-Cas system, suggests that the host possesses an active CRISPR-Cas adaptive immune system [75]. This system likely provides defence against invading genetic elements from the phages.

Biosynthetic gene cluster

The genome analysis of *S. haemolyticus* using antiSMASH revealed three biosynthetic gene clusters (BGCs) with potential roles in secondary metabolite production, which may contribute to the bacterium's competitive survival and pathogenicity (Fig. 4). The first identified BGC encodes a type III polyketide synthase (T3PKS) and terpene biosynthesis genes, suggesting the capacity for polyketide-derived metabolites and terpenoid compounds. Polyketides are structurally diverse secondary metabolites, some of which play key roles in antimicrobial activity and virulence, while terpenes are associated with bacterial communication and membrane stability [76, 77]. The second identified BGC contains genes linked to cyclic-lactone autoinducer synthesis, which is often involved in quorum sensing—a bacterial communication system that regulates gene expression, including virulence factor production [78, 79]. The presence of this cluster suggests that *S. haemolyticus* may employ quorum sensing to coordinate biofilm formation and antibiotic resistance mechanisms, further supporting its adaptability in host environments. The third BGC is responsible for non-ribosomal peptide (NRP) siderophore biosynthesis and exhibits 100% similarity to the *staphyloferrin A* gene cluster. Staphyloferrin A is an iron-chelating molecule that enhances bacterial iron acquisition, a critical function for survival in iron-limited environments such as the human host [80]. The ability to produce siderophores is a well-documented virulence factor in pathogenic *staphylococci*, allowing bacteria to outcompete host defence system by scavenging essential nutrients [81, 82]. The discovery of this BGC highlights a potential role in host colonization and infection persistence.

Genome collection and analysis

A total of 731 genome sequences of *S. haemolyticus* were fetched from the NCBI genome database. The metadata associated with each genome was fetched using Entrez E-utilities (E-utils) [51]. After filtering based on assembly quality (checkM completeness > 95%), contamination (checkM contamination < 3%), and taxonomy status (Ok), 694 sequences were selected for panresistome analysis. The sequences were submitted from 35 different countries, mostly from China (119) and USA (59). However, we couldn't find the geographic location of 239 sequences due to data labelled as 'missing' or 'not collected'. These strains were obtained from different hosts, including *Homo sapiens*, *Bos taurus*, murine, pig, canine, *Rhodotorula mucilaginosa*, and *Populus alba*, exhibiting diversity. The genome sizes ranged from 2.276 Mb (ASM2248548v1) to 2.975 Mb (ASM2622257v1). Based on PGAP annotation, the number of genes ranged between 2249 (04_NF40_HMP2135v01) and 3323 (ASM293412v2). The protein-coding genes for the selected *S. haemolyticus* genomes ranged between 2113 (ASM3575080v1) and 3077 (ASM293412v2).

Panresistome analysis

A total of 694 bacterial genomes of *S. haemolyticus* were analyzed for antimicrobial resistance (AMR) genes using the Abricate tool and three distinct databases: NCBI, ResFinder, and CARD. The comparative analysis revealed distinct and overlapping resistance gene profiles across the databases, with more than 50 resistance

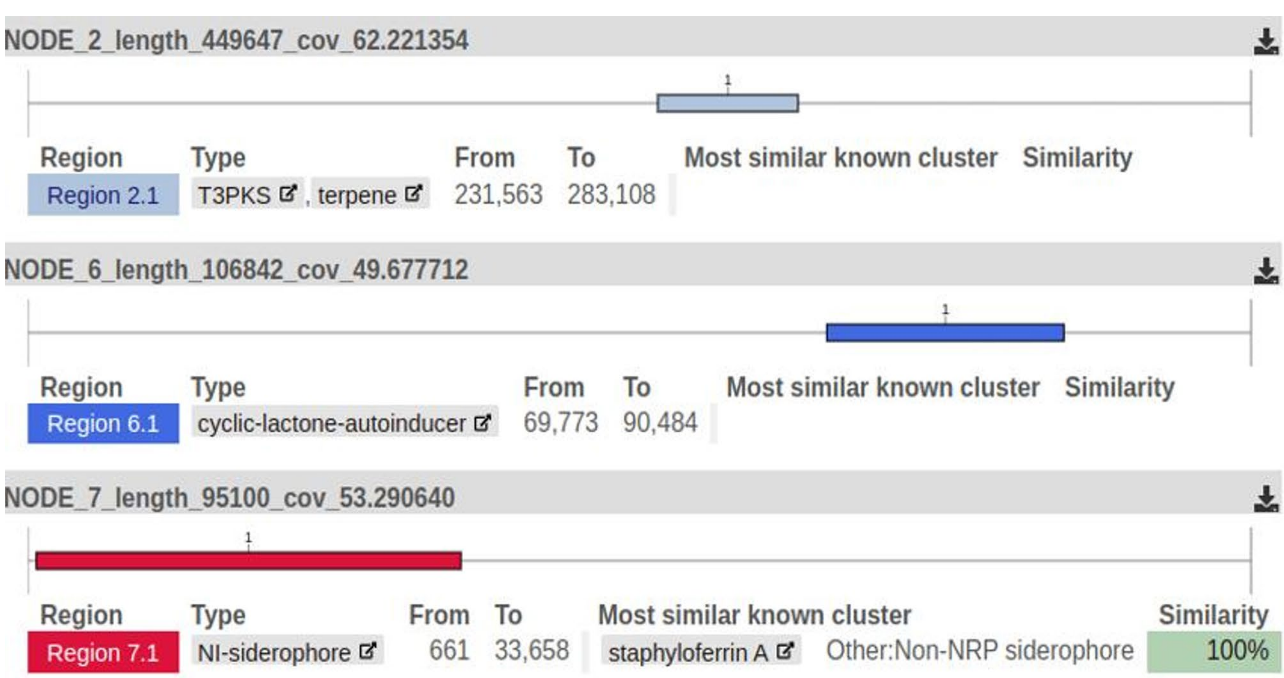

Fig. 4 Biosynthetic gene clusters of *S. haemolyticus* DUEML1

genes identified. In the NCBI database, the most prevalent resistance genes included *blaI_of_Z* (57.6%), *blaR1* (50.2%), *mecA* (48.3%), *msr(A)* (44.7%), and *mph(C)* (42.4%) (Fig. 5a). The presence of *mecA*, a well-known gene associated with methicillin resistance in *S. aureus*, was particularly notable, along with *blaR1*, which contributes to beta-lactam resistance. In the ResFinder database, *msr(A)_1* (44.6%) was the most frequently identified gene, followed by *mph(C)_2* (42.4%), *aac(6')-aph(2'')_1* (37.8%), *blaZ_78* (34.1%), and *dfrG_1* (31.3%) (Supplementary Fig. 1) (Supplementary File 1). Notably, *aac(6')-aph(2'')_1*, encoding an aminoglycoside-modifying enzyme, is a key contributor to aminoglycoside resistance. Similarly, the CARD database identified *mgrA* (80.8%) as the most prevalent gene, followed by *PC1* (49.6%), *mecA* (48.3%), *msrA* (44.7%), and *mphC* (42.4%) (Supplementary Fig. 2) (Supplementary File 1). The high prevalence of *mgrA*, a global regulator of virulence and antibiotic resistance in *staphylococci*, suggests its significant role in the resistance mechanisms of *S. haemolyticus*. The presence of *mgrA* in the *S. haemolyticus* ST-184 reflects its similarities to the broader *S. haemolyticus* population. The analysis also highlighted the diversity of resistance mechanisms, including beta-lactam resistance (*mecA*, *blaZ*), aminoglycoside resistance (*aac(6')-aph(2'')_1*, *aph(3')-IIIa*), macrolide resistance (*msr(A)*, *mph(C)*), and tetracycline resistance (*tet(K)*, *tet(M)*).

Furthermore, the prevalence of these AMR genes (prevalence > 10%) was analyzed comparatively across different geographic locations where the number of submitted sequences were at least five. The comparative analysis of resistance gene prevalence across different regions reveals distinct geographical patterns in *S. haemolyticus* genomes (Fig. 6, Supplementary Fig. 3 and Fig. 4) (Supplementary File 1). Across all three databases—NCBI, ResFinder, and CARD—France emerging as the most concerning hotspot showed universal resistance to critical antibiotics (100% prevalence of *mecA*, *aph(2'')-Ii*, and *msr(A)*), followed closely by Egypt (100% *mecA*, *blaPC1*, and *blaR1*; 90% *aph* genes) and Nigeria

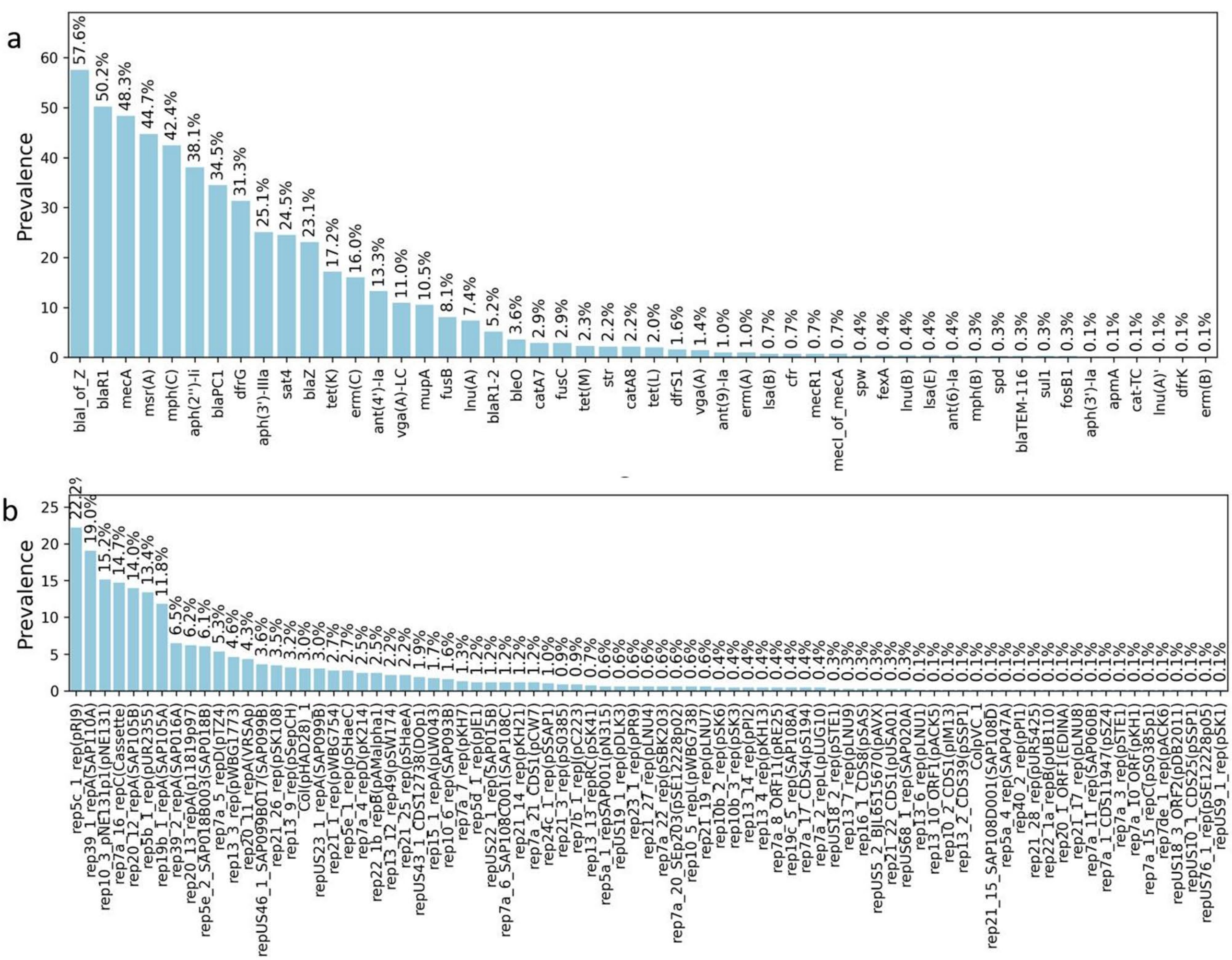

Fig. 5 Prevalence of (**a**) antibiotic resistance genes (NCBI database) and prevalence of (**b**) plasmid-associated genes (PlasmidFinder database)

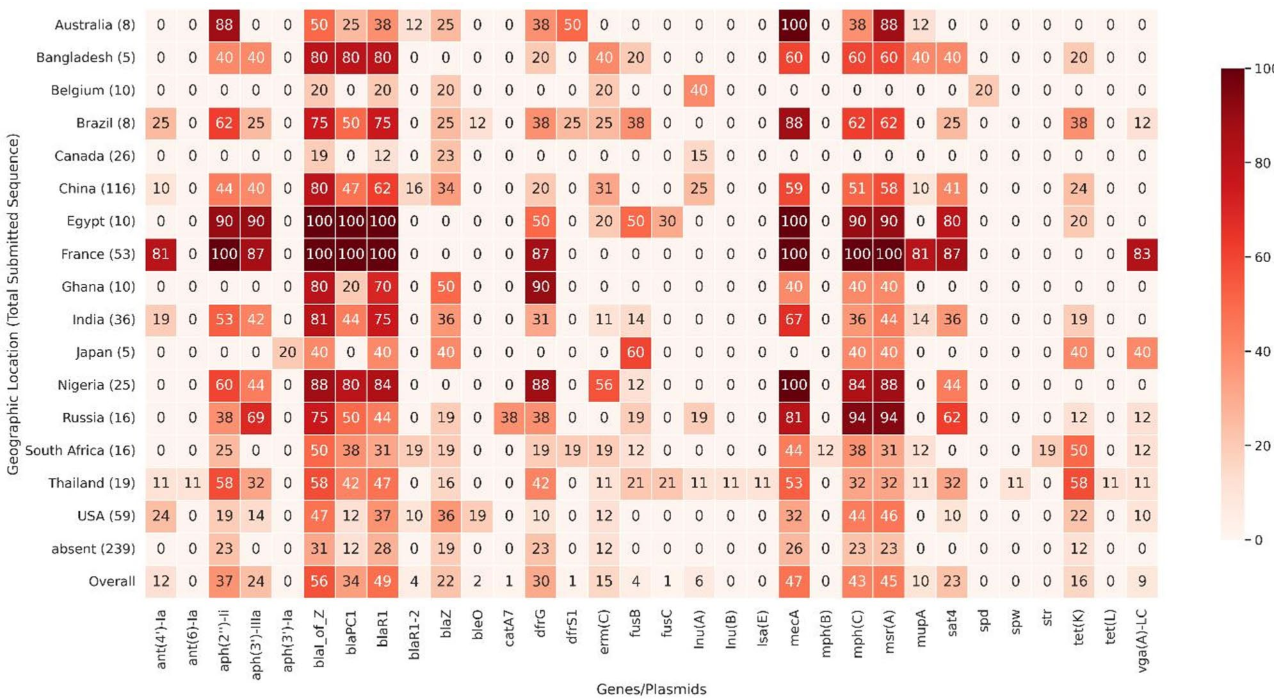

Fig. 6 Comparative analysis of the prevalence of antibiotic resistance genes (NCBI database) across different geographic locations (No. of sequences > 5)

(100% *mecA*, 88% *msr(A)*, 84% *mph(C)*), indicating severe multidrug resistance patterns likely driven by extensive antibiotic use and nosocomial transmission. Bangladesh presented an alarming resistance profile with 60% *mecA* prevalence, 80% occurrence of β-lactamase regulators (*blaI_of_Z*, *blaPC1*, *blaR1*), and 60% macrolide resistance (*msr(A)*, *mph(C)*), positioning it alongside other high-burden Asian nations like China (58.6% *mecA*, 43.9% *aph(2")-Ii*) and India (66.6% *mecA*, 44.4% *msr(A)*), while Brazil showed exceptionally high β-lactam resistance (87.5% *mecA*) coupled with significant aminoglycoside resistance (62.5% *aph(2")-Ii*). Russia demonstrated unique resistance dynamics with extreme macrolide resistance (93.75% *mph(C)* and *msr(A)*) alongside high *mecA* prevalence (81.25%), whereas Thailand exhibited worrying tetracycline resistance (57.8% *tet(K)*) and moderate methicillin resistance (52.6% *mecA*). South Africa presented a divergent pattern with high tetracycline (50% *tet(K)*) but lower β-lactam resistance (43.75% *mecA*), while Ghana showed unexpectedly high trimethoprim resistance (90% *dfrG*) despite lower *mecA* prevalence (40%). Belgium's profile was notable for complete absence of *mecA* but presence of other resistance genes (20% *erm(C)*, 40% *lnu(A)*), and Australia displayed selective resistance with 87.5% *aph(2")-Ii* but no *erm* genes. The United States showed moderate resistance across multiple classes (32.2% *mecA*, 45.7% *msr(A)*), while Canada and Japan stood out as low-resistance regions with no *mecA* detected, though Japan showed localized fusidic acid resistance (60% *fusB*). Thailand's resistance landscape was particularly complex with high *tet(K)* (57.8%) and emerging β-lactam resistance (52.6% *mecA*), and Ghana's extremely high *dfrG* (90%) suggested specific antibiotic selection pressures. These findings collectively demonstrate that while some countries (France, Egypt, Nigeria) face near-pan-resistant *S. haemolyticus* strains, others (Bangladesh, China, Brazil) show emerging multidrug resistance with distinct genetic patterns, and a third group (Canada, Japan, Belgium) maintains relatively controlled resistance profiles. These variations highlight global differences in antibiotic usage, selective pressures, sequencing biases, and resistance dissemination across *S. haemolyticus* populations. These findings also highlight the urgent need for region-specific intervention strategies to combat the rising threat of MDR *staphylococcal* infections worldwide.

Comparative analysis of plasmids and virulence factors

All 694 *S. haemolyticus* genomes were analyzed for virulence factors and plasmids. The screening of virulence factors using VFDB identified three distinct virulence factor genes within the *S. haemolyticus* genomes, with notable geographic variability (Fig. 7). The most prevalent virulence genes were *cap8G*, detected in 117 genomes (16.88%), and *cap8E*, present in 104 genomes (15.01%). Both genes are integral to the capsule polysaccharide biosynthesis pathway, a key mechanism for immune evasion that enables bacteria to resist host defenses [83]. The relatively high prevalence of these genes suggests that capsule formation is an important virulence strategy

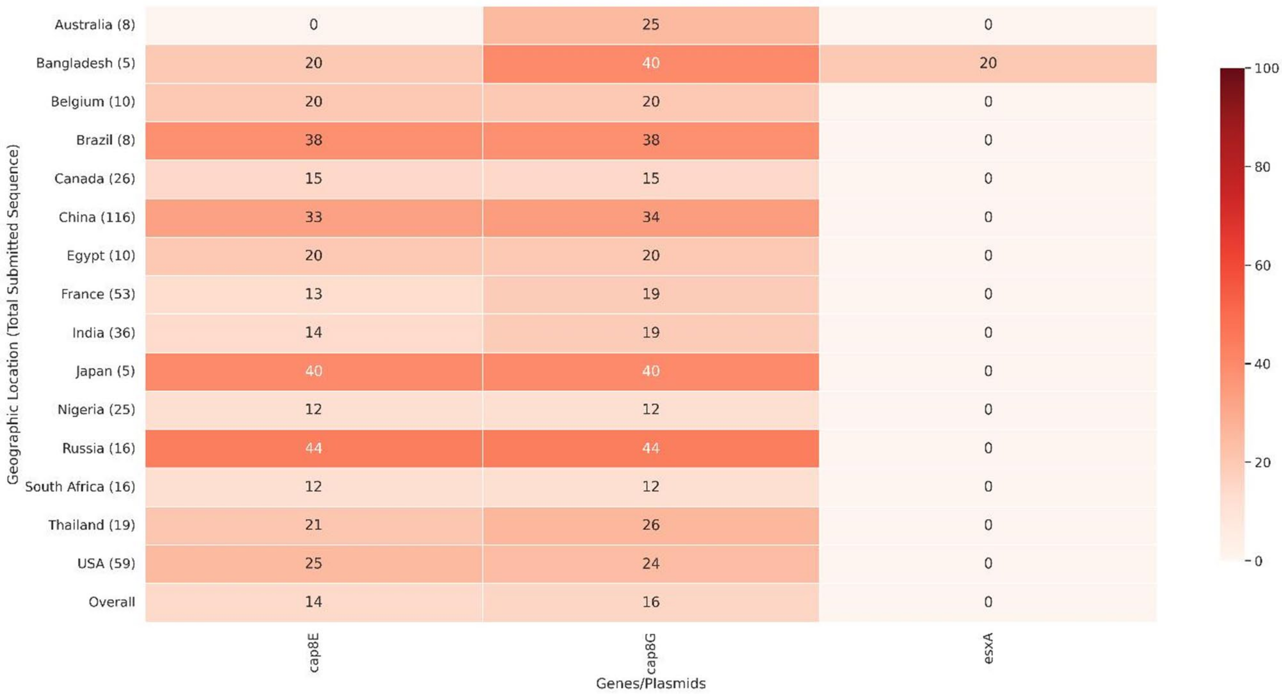

Fig. 7 Comparative analysis of the prevalence of virulence genes (VFDB database) across different geographic locations

in *S. haemolyticus*. In contrast, *esxA*, a gene associated with the ESAT-6 secretion system, was found in only 2 genomes (0.29%). One of these was the ST-184 *S. haemolyticus* strain characterized in this study. The second *esxA*-positive genome (ASM514626v2) was also derived from *Homo sapiens* but lacks associated metadata and has not been described in prior literature. This suggests a lack of genomic analysis even in the era of widespread sequencing, revealing limitations in current pathogen surveillance systems.

Geographic comparisons revealed distinct patterns in virulence factor distribution. The *cap8E* gene was most prevalent in Russia (44%) and Brazil (38%), whereas it was entirely absent in Australia. The *cap8G* gene displayed a broader distribution, with the highest frequencies observed in Russia (44%), Japan (40%), and Bangladesh (40%), but lower occurrences in Nigeria (12%) and Australia (25%). Overall, Russia and Japan exhibited the highest prevalence of *cap8E* and *cap8G*, reinforcing the importance of capsule biosynthesis in these regions, while Bangladesh was distinct for its detection of *esxA*. These findings underscore the geographic diversity in virulence factor distribution among *S. haemolyticus* isolates, likely shaped by regional differences in ecological and evolutionary pressures.

The screening of plasmid-associated genes using the PlasmidFinder database revealed their widespread presence in *S. haemolyticus* genomes. A total of 81 different plasmid-associated genes were identified across the genomes, suggesting a high degree of genetic adaptability facilitated by horizontal gene transfer. The most frequently detected plasmid gene was *rep5c_1_rep* (pRJ9), present in 154 genomes (22.2%), followed by *rep39_1_repA* (SAP110A) (19.0%), *rep10_3_pNE131p1* (pNE131) (15.2%), *rep7a_16_repC* (Cassette) (14.7%), and *rep20_12_repA* (SAP105B) (14.0%) (Fig. 5b). The high prevalence of these plasmid-associated genes indicates their potential role in the dissemination of antimicrobial resistance and virulence factors within *S. haemolyticus* populations. Several of the identified plasmids have known associations with antibiotic resistance. For instance, *rep39_1_repA* (SAP110A) and *rep20_12_repA* (SAP105B) have been linked to resistance determinants for aminoglycosides and β-lactams, which are critical in clinical settings [84]. Additionally, *rep5b_1_rep* (pUR2355) has been associated with glycopeptide resistance [85]. The presence of these resistance-associated plasmids underscores the importance of monitoring plasmid-mediated resistance in *S. haemolyticus* infections, as they can significantly impact the effectiveness of therapeutic interventions.

Similar to resistance genes, the prevalence of plasmid-associated genes (prevalence > 10%) was analyzed comparatively across different geographic locations where the number of submitted sequences were at least five. The comparative analysis of plasmid genes distribution across different geographic regions revealed notable regional variations (Fig. 8). In China, *rep39_1_repA* (SAP110A) was the most prevalent plasmid-associated gene, present in 46.55% of the genomes, while *rep10_3_pNE131p1*

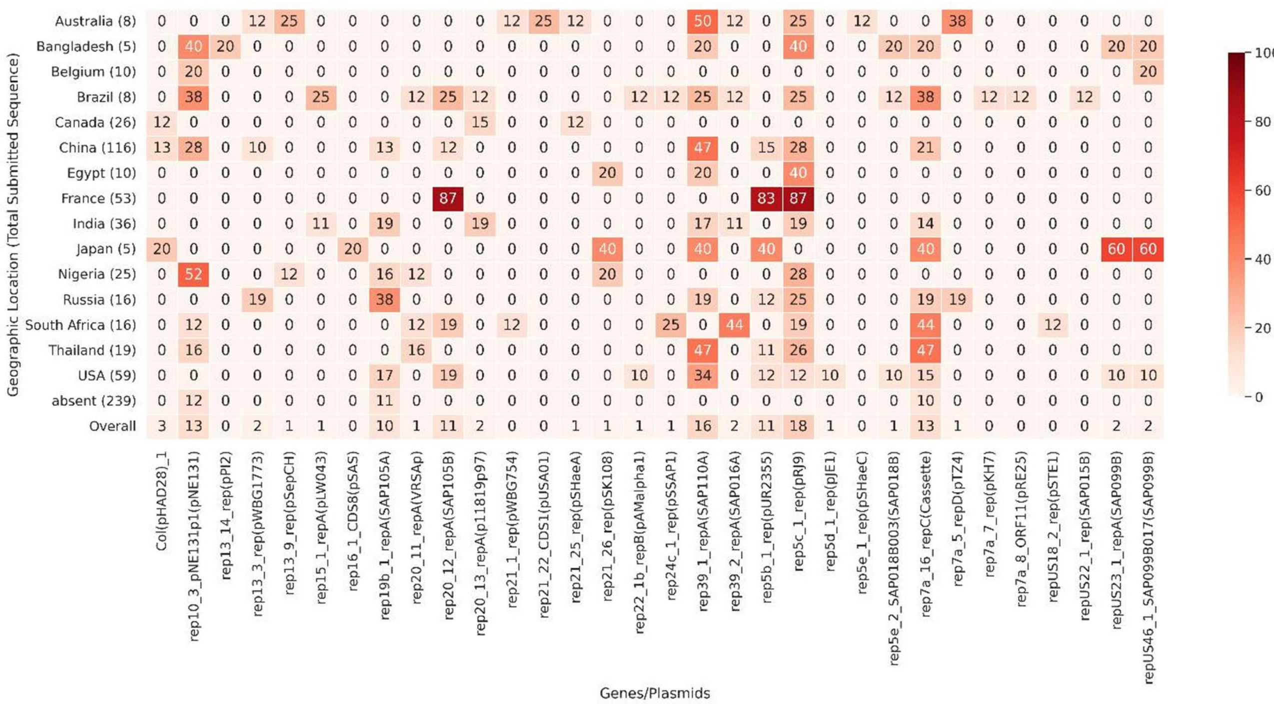

Fig. 8 Comparative analysis of the prevalence of plasmid-associated genes (PlasmidFinder database) across different geographic locations (No. of sequences > 5)

(pNE131) and *rep5c_1_rep* (pRJ9) were also highly prevalent at 28.45% each. In contrast, France exhibited a strikingly high prevalence of *rep20_12_repA* (SAP105B) and *rep5c_1_rep* (pRJ9), both at 86.79%, and *rep5b_1_rep* (pUR2355) at 83.02%, indicating a distinct plasmid profile. The USA showed moderate prevalence of *rep39_1_repA* (SAP110A) at 33.90%, with *rep20_12_repA* (SAP105B) and *rep5c_1_rep* (pRJ9) at 18.64% and 11.86%, respectively. Nigeria displayed a high prevalence of *rep10_3_pNE131p1* (pNE131) at 52.00%, with *rep5c_1_rep* (pRJ9) at 28.00%. In Japan, *repUS23_1_repA* (SAP099B) and *repUS46_1_SAP099B017* (SAP099B) were notably prevalent at 60.00%, while *rep39_1_repA* (SAP110A) was present in 40.00% of the genomes. South Africa had a high prevalence of *rep39_2_repA* (SAP016A) at 43.75% and *rep7a_16_repC* (Cassette) at 43.75%. Thailand exhibited a significant presence of *rep39_1_repA* (SAP110A) at 47.37% and *rep7a_16_repC* (Cassette) at 47.37%. In Bangladesh, *rep10_3_pNE131p1* (pNE131) and *rep5c_1_rep* (pRJ9) were both prevalent at 40.00%. In contrast, Australia and Canada had relatively lower overall plasmid prevalence, with *rep39_1_repA* (SAP110A) at 50.00% in Australia and *rep20_13_repA* (p11819p97) at 15.38% in Canada. These findings highlight the significant geographic variability in distribution of plasmid-associated genes among *S. haemolyticus* isolates. The high prevalence of certain plasmids in specific regions suggests regional differences in genetic repertoire likely influenced by selective pressures such as antibiotic use and horizontal gene transfer dynamics.

The diverse repertoire of plasmid-associated genes, antibiotic resistance genes, and virulence genes explored in the study is not comprehensive since some regions have fewer sequence reports. This limitation suggests that the current dataset may not fully represent the global genetic diversity of *S. haemolyticus*. Underrepresentation of certain geographic locations could lead to an incomplete understanding of the distribution of resistance and virulence factors. Expanding sequencing efforts in under-reported regions would provide a more accurate and holistic view of the genomic characteristics of *S. haemolyticus* worldwide.

Conclusion

In this study, the MDR DUEML1 (ST184) collected from a respiratory tract infection was meticulously studied. Using whole-genome sequencing and comprehensive genome analysis, the intricate genetic landscape of this isolate was revealed. The isolate harbored multiple antimicrobial resistance genes, highlighting mechanisms underlying multidrug resistance, including the role of MGEs in facilitating horizontal gene transfer. These MGEs included phage-associated elements, recombination- and replication-related genes, and competence-related genes, which likely contribute to the acquisition and dissemination of resistance determinants and virulence factors. For example, phage-associated genes suggest phage-driven gene mobilization, while recombination-related genes indicate active genomic rearrangements enhancing adaptability. Competence-associated genes point to the ability of ST184 to acquire extracellular DNA, supporting genetic exchange. Alien Hunter analysis further predicted several potential HGT events, underscoring the dynamic nature of this genome in acquiring virulence and resistance traits. The study also uncovered CRISPR-associated sequences, plasmids, and insertion sequences, reflecting the isolate's genetic diversity and capacity to mobilize resistance traits. Comparative analysis of 694 *S. haemolyticus* genomes revealed a diverse array of plasmids, virulence factors, and resistance genes across isolates from 35 different countries, highlighting the global significance of this pathogen as a contributor to antimicrobial resistance. The strength of this study lies in being the first genomic characterization of an *esxA*-positive *S. haemolyticus* isolate from Bangladesh, where the discovery of a novel sequence type (ST-184) and comprehensive genomic analyses provide valuable insights into its resistance, virulence, and global significance. These findings can serve as reference for subsequent studies at a global scale. Since this represents one of the few genomic analyses of *S. haemolyticus* in Bangladesh, further investigations are warranted. Future work should employ comprehensive whole-genome sequencing across larger collections of isolates to thoroughly resolve the genomic architecture of *S. haemolyticus*, including the detection of silent resistance genes embedded within its genome. These investigations should focus on unraveling the bacterium's pathogenic potential and the complex mechanisms driving its resistance traits. Additionally, a concerted effort is needed to develop innovative and effective therapeutic strategies to address the challenges posed by this MDR pathogen.

Abbreviations

MDR	Multi-drug resistant
CLSI	Clinical & Laboratory Standards Institute
OD	Optical Density
NCBI	National Center for Biotechnology Information
BGC	Biosynthetic gene clusters
AMR	Antimicrobial resistance

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12866-025-04406-5.

Supplementary Material 1.

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Authors' contributions
TAA designed the study and analyzed the data under the supervision of AB. TAA wrote the first draft of the manuscript with significant contributions from FI and RU. FI, TAA and RR curated and pre-processed the datasets. AB was the principle investigator and reviewed the manuscript with HA and SAB. All authors read and approved the final manuscript.

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Data availability
The whole genome sequence of *Staphylococcus haemolyticus* DUEML1 was deposited in DDBJ/ENA/GenBank of NCBI under BioProject PRJNA1215361, BioSample SAMN46402922 and Accession no. JBLFEB000000000.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Ethical approval for this study was obtained on May 12, 2025, from the Ethical Review Committee of the Faculty of Biological Sciences, University of Dhaka (Ref. No. 290/Biol.Scs/). All methods were carried out in accordance with relevant guidelines and regulations. Informed consent was obtained from the patient for sample collection, research conduct, and publication of the findings.

Consent for publication
Not Applicable.

Competing interests
The authors declare no competing interests.

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